

PROJECT PROPOSAL

BLOCK 1D-2023-2024

SmartAccidentPredict

Names of the Students

Kees Klijs

Jędrzej Ludwiczak

Michał Bątkowski

Viktória Kubišová

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I. Introduction

The objective of the SmartAccidentPredict project is to develop an advanced dashboard that integrates a machine learning model to predict the severity of traffic accidents based on specific accident details. This initiative aims to enhance road safety in Breda by providing actionable insights and enabling efficient mobilization of law enforcement and emergency services. Additionally, the dashboard will identify and analyze the most dangerous intersections within the city, using clustering techniques to uncover patterns and contributing factors to accidents in these areas.

The relevance of this project stems from the critical need to improve road safety and reduce accident-related fatalities and injuries. Breda, like many urban areas, faces challenges related to traffic management and accident prevention. By leveraging machine learning, we can create a predictive model that not only forecasts the severity of accidents but also provides data-driven insights to city planners and traffic management authorities. This proactive approach can lead to better resource allocation, timely incident responses, and informed traffic infrastructure decisions, potentially saving lives and minimizing damage. It benefits the municipality by improving community well-being and quality of life while ensuring efficient use of resources and significant cost savings, allowing funds to be reallocated to other projects.

The application of machine learning techniques is significant in this context because it allows for the analysis of large and complex datasets, uncovering patterns that might not be immediately apparent through traditional analysis methods. Machine learning models can process real-time data from cameras and sensors placed around the city, providing continuous monitoring and predictive capabilities. This technology-driven approach ensures that safety measures are both responsive and preventative, ultimately contributing to a safer driving environment in Breda.

II. Problem Statement

Problem Definition

The current problem facing law enforcement and emergency services is that they often arrive too late at the scene of a traffic accident, which could have been prevented or mitigated with the introduction of machine learning techniques. This delay not only affects the quality of services provided by local authorities but also impacts the safety and well-being of the community. In severe incidents, delayed responses can mean the difference between life and death. Specifically, the problem we aim to address with SmartAccidentPredict is the prediction and mitigation of accident severity in Breda using machine learning techniques. We seek to develop a predictive model that can accurately assess the severity of accidents based on real-time data collected from cameras and sensors deployed across the city. This model will classify accidents into categories of severity (Fatal, Injured, Material damage only) using features such as road details, date of occurrence, weather conditions, and vehicle characteristics.

Context and Background

Breda, a bustling city in the Netherlands, faces ongoing challenges related to road safety despite continuous efforts by organizations like ANWB to promote safe driving practices. Traffic accidents not only disrupt daily life but also pose significant risks to public health and safety. The city has invested significantly in road infrastructure and safety campaigns, yet accidents continue to occur, highlighting the need for more proactive and predictive measures. Current methods of accident response and prevention can be enhanced through predictive analytics, enabling authorities to deploy resources more effectively and implement targeted safety measures.

Solution

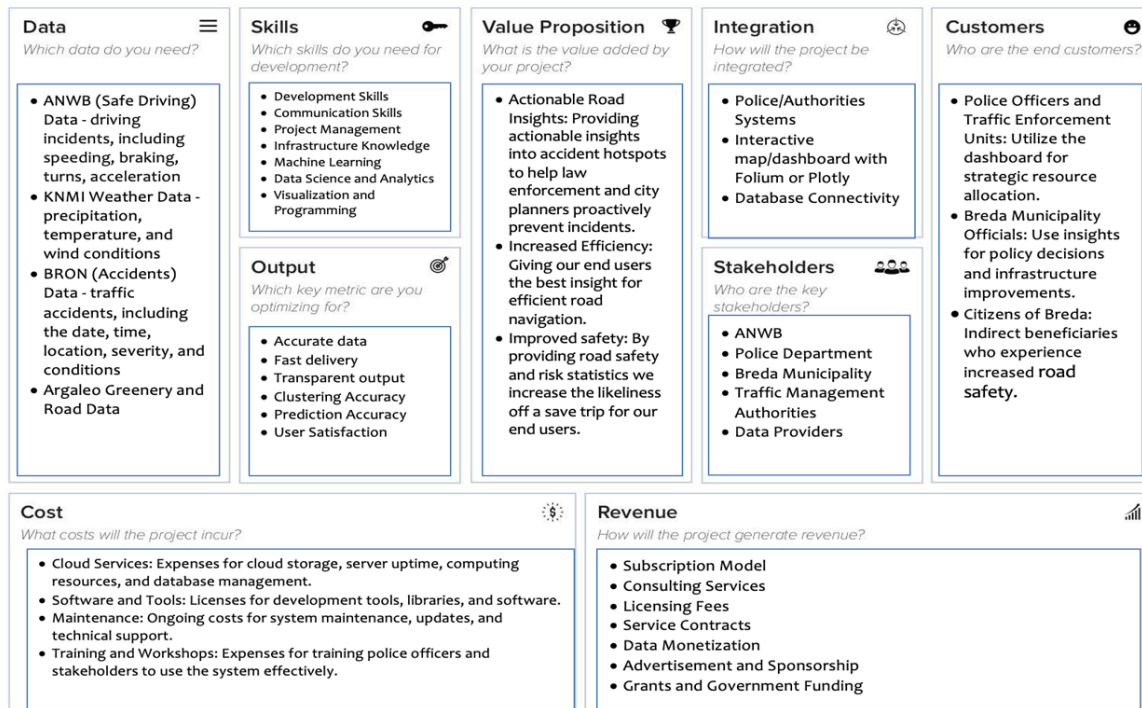
Solving the problem of accident severity prediction through machine learning offers significant benefits. Machine learning facilitates data-driven decision-making by uncovering hidden patterns from large datasets, informing targeted traffic management strategies and infrastructure improvements. The primary goals of the SmartAccidentPredict project are:

1. **Develop a Predictive Model:** Create a machine learning model capable of predicting accident severity based on real-time data inputs. This involves analyzing data from ANWB and other relevant sources to train algorithms that can foresee the occurrence and severity of accidents.
2. **User-Friendly Dashboard:** Implement the model into a user-friendly dashboard accessible to ANWB, city officials, and emergency services for real-time monitoring and decision-making. This dashboard will demonstrate what the algorithm could do in real time if applied, enhancing the decision-making process for law enforcement and emergency services.
3. **High-Risk Intersections Analysis:** Identify and analyze high-risk intersections in Breda using clustering techniques to understand contributing factors to accidents in these areas. This analysis will help in providing actionable insights to stakeholders to support targeted interventions and infrastructure improvements.
4. **Targeted Interventions:** Provide actionable insights to stakeholders to support targeted interventions and infrastructure improvements aimed at reducing accidents and improving road safety. By predicting accident severity, authorities can prioritize emergency responses and allocate resources more efficiently, thus reducing response times and minimizing accident impacts.

Proactive accident prevention not only reduces healthcare costs, vehicle repair expenses, and insurance claims but also enhances the overall quality of life in Breda by fostering a safer environment for residents and visitors alike. The SmartAccidentPredict project aims to bring a significant positive impact on road safety and emergency response efficiency in Breda.

AI Project Canvas

Title: **Accident Hotspot Identification Dashboard**



III. Data Description

For this project, we plan to use all the available datasets from the provided data lake, including ANWB (Driver Characteristics), KNMI (Weather), BRON (Accidents), and Argaleo (Greenery/Road). Each of these datasets offers a wealth of information critical for our analysis, particularly for clustering to understand the patterns and factors contributing to road safety incidents.

The ANWB dataset provides comprehensive details about driving incidents, such as the event's start and end times, location coordinates, speed measurements, and the type and severity of the incident. Key features for clustering from this dataset include "duration_seconds", "latitude", "longitude", "speed_kmh", "end_speed_kmh", "maxwaarde", "category", and "incident_severity". These attributes help in understanding the spatial and temporal distribution of incidents, as well as the driving behaviors associated with different types of incidents.

From the KNMI datasets, we will use weather-related variables, including precipitation, temperature, and wind data. Important features include "precipitation intensity", "air temperature", and "wind speed". These attributes are crucial for correlating weather conditions with driving incidents, helping us to determine if certain weather patterns contribute to higher risks of accidents.

The BRON dataset on accidents contains detailed information about each accident, such as the date and time, location, number of injuries or fatalities, and road conditions. Key features here include "date", "time", "latitude", "longitude", "number_of_injuries", and "road_conditions". This data allows us to analyze accident hotspots and the severity of accidents under varying conditions.

Lastly, the Argaleo datasets provide information on road characteristics and greenery, with features such as "road segment id", "road number", "road name", and "relative height". These attributes are essential for understanding how the physical and environmental characteristics of roads influence accident occurrences.

As the BRON dataset on accidents initially intended for accident severity prediction lacks a lot of information and most of the columns are empty, we will utilize the Accidents_17_23 dataset provided by a colleague, which includes comprehensive information such as "Year", "Accident severity", "Town", "First mode of transport", "Second mode of transport", "Area type", "Light condition", "Road location", "Road condition", "Road surface", "Road situation", "Speed limit", "Street", "Weather", and "Accidents". This dataset includes sufficient information to develop a model for accident severity prediction.

While these datasets are comprehensive, much of the additional data from other sources had significant missing values. Therefore, we will mainly focus on the Accidents_17_23 dataset because the original accidents dataset lacked sufficient information. To enhance the analysis, we created our own dataset for intersection accidents clustering, with significant contributions from Kees.

In the context of the business setting, the city of Breda's existing infrastructure of cameras, speedometers, and sensors would provide real-time data to enhance our predictive model as well as make real-time predictions. This integrated approach aims to improve road safety by accurately forecasting accident hotspots in specific areas. The Streamlit interactive dashboard, which is to be developed, is meant to showcase what the model can do with example inputs.

IV. Methodology

(In this section, you should outline the approach you plan to take to tackle the problem. Describe the steps involved in the machine learning process, including data collection, pre-processing, model selection, training, evaluation, and deployment. Furthermore, mention any specific techniques, algorithms, or tools you intend to use.)

Briefly describe the machine learning model(s) you intend to build. Explain why you chose these models and how they are suitable for addressing the problem. If possible, provide an overview of the model architecture or design.

Explain the level of risk associated with AI system you intend to build. Provide a list of associated legal obligations that must be followed and explain your approach for addressing them.)

Our approach to tackling the problem of accident severity prediction and intersection clustering for road safety in Breda involves several key steps within the machine learning process. Initially, we will gather and integrate relevant datasets from ANWB (Driver Characteristics), KNMI (Weather), BRON (Accidents - focusing on Accidents_17_23), and Argaleo (Greenery/Road). These datasets will undergo thorough preprocessing to clean, normalize, and integrate them into a unified format suitable for analysis.

Data Collection: The primary datasets for this project are provided by the Dutch insurance company ANWB, focusing on "Accidents" and "Safe Driving" datasets. These datasets include detailed information about traffic incidents, driver behavior, and environmental conditions. The Accidents_17_23 dataset will also be used because the original accidents

dataset lacked sufficient information. Additional data from KNMI, BRON, and Argaleo, covering weather and road characteristics, will also be used if relevant and complete. This diverse data is crucial for building accurate predictive models.

Data Pre-processing: In the data pre-processing phase, we will clean the data by handling missing values, removing duplicates, and correcting inconsistencies to ensure quality and reliability. We will then split the data into training, validation, and test sets to evaluate model performance accurately and prevent overfitting.

Feature Engineering: In this phase we will extract relevant features such as accident location, weather conditions, road characteristics, and driving behavior metrics from ANWB data to enhance the model's predictive capabilities.

Model Selection: This phase involves choosing appropriate machine learning models for accident severity prediction and intersection clustering. For accident severity prediction, we will consider supervised learning models such as Logistic Regression, Random Forest or Gradient Boosting due to their ability to handle complex relationships in data and provide interpretable results. For intersection clustering, unsupervised learning techniques like K-means and DBSCAN clustering will be employed to identify accident-prone areas.

Model Training:

Each model will undergo a thorough training process where data will be preprocessed separately. To address class imbalance in accident severity data, techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or assigning class weights will be used. Hyperparameter tuning will be conducted in the final iteration to optimize model performance.

- *Logistic Regression, Random Forest, Gradient Boosting:*
 - Data will be balanced using the SMOTE method or class weights.
 - Models will be trained on the processed dataset.
 - Hyperparameter tuning will be performed to enhance model performance.
- *Convolutional Neural Network (CNN):*
 - Suitable preprocessing techniques will be applied.
 - The model will be trained on spatial data to capture geographical patterns.
 - Hyperparameter tuning and data augmentation techniques will be used to prevent overfitting.

Model Evaluation:

Model performance will be evaluated using standard metrics such as accuracy, precision, recall, F1 score, and ROC-AUC for classification models. For clustering models, evaluation metrics will include silhouette score and Davies-Bouldin index. Cross-validation techniques will be applied to ensure robust model performance.

Model Deployment:

The best-performing model will be integrated into the Streamlit interactive dashboard. This dashboard will showcase all visuals and data analysis results, allowing users to input data

and receive real-time predictions. The model will process user inputs and provide actionable insights to law enforcement and emergency services.

- *Integration into Dashboard:*
 - The final model will be deployed within the Streamlit framework.
 - Users will interact with the dashboard to input data and view predictions.
 - Real-time data processing capabilities will be included to enhance usability and effectiveness.

Legal Obligations and Approach:

High-Risk Classification (EU AI Act Articles 6-15):

Reason: Predicts incidents influencing safety measures and law enforcement actions.

Transparency (Article 52):

Inform users that predictions are AI-generated to foster trust and understanding.

Post-Market Monitoring (Articles 61-62):

Continuous monitoring to ensure ongoing compliance, safety, and reliability after deployment. Regular updates to maintain accuracy and prevent bias.

Prohibited Practices (Article 5):

Avoid exploiting vulnerabilities, using subliminal techniques, or engaging in social scoring.

Compliance Measures:

- Maintain detailed documentation of AI system development, data sources, and decision-making processes.
- Provide training to users on AI system capabilities and limitations.
- Implement continuous monitoring to regularly update and assess the system for accuracy, bias, and potential harm.

V. Project Timeline

(In this section, you should break down the project into specific tasks and assign a timeframe to each task. Provide a timeline that outlines when each task will be completed, from data pre-processing to model deployment. Additionally, you need to document the individual contributions of each team member.)

Weeks 1 to 3: Preparation and Learning During these initial weeks, each team member will focus on understanding the legal framework, developing a comprehensive business understanding relevant to the project, and gaining knowledge about manipulating data series to analyze datasets with dates. This foundational work is crucial to ensure that the project complies with legal requirements, aligns with business objectives, and effectively handles time-based data. This phase of the project was individual.

Week 3: Project Planning and Data Collection: This week, the team will establish clear goals and assign specific tasks. Data collection from ANWB and other sources will commence, integrating these datasets into a unified repository. In lieu of meetings, the team will focus on developing the project roadmap, generating project ideas, and conducting initial research. Additionally, efforts will be made to enhance business understanding and apply the AI Canvas model to the project. Regular documentation of progress and status reports will continue. This was mainly done by Kees, Michal and Viktoria.

Week 4: Data Pre-processing and EDA This week involves cleaning and preparing the data by addressing missing values and inconsistencies. The team will conduct exploratory data analysis (EDA) to understand data distributions and identify patterns. Additionally, the initial design and development of the interactive dashboard will begin, supported by continuous team meetings for smooth collaboration and thorough documentation. This was mainly done by Kees and Michal.

Week 5: Model Development and Initial Dashboard Integration The focus will shift to selecting and developing machine learning models. The team will train and evaluate various models, optimizing their performance. Simultaneously, the dashboard development will progress with the integration of initial model outputs. Regular updates and coordination will be maintained through team meetings. This was mostly done by Jędrzej and Viktoria.

Week 6: Model Deployment and Testing The trained model will be deployed into the interactive dashboard for real-time predictions. The team will conduct thorough testing and validation to ensure the system's accuracy and reliability. Based on user feedback, further refinements will be made to the dashboard. Ongoing team meetings will facilitate final preparations and coordination. This was mostly done by Jędrzej.

Week 7: Review, Optimization, and Finalization In the final week, the team will review the entire project, optimizing both the model and the dashboard performance. Final touches will be added to the dashboard to prepare it for presentation. Comprehensive documentation and final reports will be prepared, ensuring all project deliverables are polished and complete. During this phase all of the team members were participating.